Celebrity identification and detection

IE 7615 | Project 1 | Group 1

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Introduction and background

This project examined how deep learning could identify celebrities in individual photographs and locate them in group images. Classification assigned an identity to a single face. Detection added bounding boxes so several faces could be labelled in one image.

Three classification models were compared before a synthetic group-image dataset was built and YOLOv8n was fine-tuned. The final system combined the detector with ResNet18 to return face boxes and identity labels. Withholding uncertain labels reduced wrong identities, but also left more known faces unnamed.

Dataset construction

Five CelebA identities were used throughout: 7007, 2970, 2336, 7 and 4428. Classification used 23 photographs per identity. Detection used all 121 available photographs, preserving the earlier split assignments and adding the six unused photographs to validation and test.

Each synthetic scene combined two to five identities on a drawn room background at 800 by 600 pixels. Position, scale and brightness varied. YuNet supplied face boxes, and the source-photo records supplied the identities. Labels were stored in YOLO format with coordinates normalized to the image dimensions.

Dataset	Train	Validation	Test
Classification photographs	85	15	15
Detection source photographs	85	19	17
Original synthetic scenes	168	36	36
Final detection scenes	504	36	36

Source photographs were split before composition to prevent reuse across splits. The original scenes followed a 70/15/15 split. Two augmented copies of each training scene brought the dataset to 576 images and 1,820 labelled faces. Validation and test received no extra copies.

Training augmentation included horizontal flips, small geometric changes, lighting and colour changes, blur, noise and compression. Images and boxes were transformed together. File pairs, box bounds, class coverage and shared source photographs were checked before training.

Guide to the report and project

Classification is on page 2, detection on page 3, the combined pipeline on page 4, the gallery on page 5, limitations on page 6 and ethics on page 7. The Milestone 4 repository guide provides setup and reproduction instructions. The app ran with `python project_milestone04/demo.py`; batch inference ran with `python project_milestone04/pipeline.py image.jpg`.

Classification architecture and results

The small CNN used two convolution blocks with 32 and 64 channels. The deeper CNN added blocks with 128 and 256 channels. Both used ReLU, max pooling, global average pooling and dropout before a five-class output layer. ResNet18 instead kept its ImageNet-pretrained feature layers fixed and trained only a new five-class output layer.

All models received 224 by 224 images with ImageNet normalization. Random crops and horizontal flips were applied during training; validation and test preprocessing stayed fixed. Training ran for 30 epochs on CPU with Adam, learning rate 0.001, batch size 16 and seed 47.

Each checkpoint was selected by its lowest validation loss. ResNet18 was then chosen by validation accuracy before the test results were examined. All three models were tested on the same 15 photographs, with three per identity.

Model	Trainable parameters	Training time	Test accuracy
Small CNN	19,717	92 s	33.3%
Deeper CNN	389,701	146 s	40.0%
ResNet18	2,565	116 s	66.7%

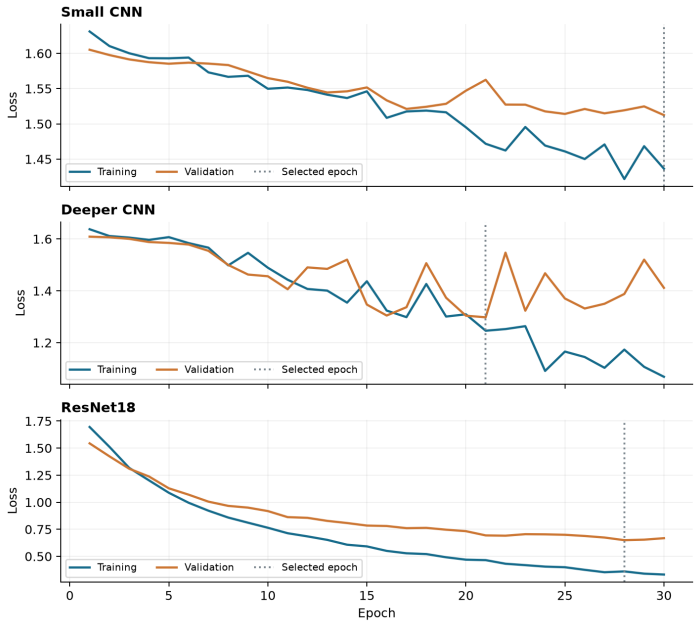


Figure 1. Recorded training and validation losses. Dotted lines mark the saved checkpoints.

ResNet18 performed best, suggesting that pretrained features helped with the small dataset. The deeper CNN improved on the small CNN by only one test photograph. Extra layers alone did not close the gap, and the small test set limited how strongly these differences could be interpreted.

ResNet18 classified all test photos for identities 2336 and 7 correctly, but only one of three for both 7007 and 4428. Overall accuracy therefore hid differences between identities.

Detection training and results

COCO-pretrained YOLOv8n was fine-tuned for 50 epochs at image size 640 and batch size 8, using seed 42. The automatic optimizer selected AdamW with an initial learning rate of 0.001111. Training took about eight minutes on a Tesla T4. Epoch 29 was saved because it had the best validation mAP across IoU thresholds from 0.50 to 0.95.

Evaluation used 36 held-out scenes containing 113 faces. A correct match required the right identity and box overlap, measured by intersection over union (IoU), of at least 0.50.

Metric	Baseline	Selected settings
mAP at IoU 0.50	0.962	0.962
mAP across IoU 0.50 to 0.95	0.862	0.862
Precision	0.898	0.929
Recall	0.938	0.920
Mean IoU of correct matches	0.931	0.932

Precision measured how many predictions were correct; recall measured how many labelled faces were correctly found. Mean average precision (mAP) summarized confidence cutoffs. Mean IoU included correct matches only, excluding missed or wrongly identified faces.

The selected cutoff retained fewer wrong predictions but missed more correct ones. The mAP scores stayed unchanged because both settings used the same checkpoint and duplicate filtering; mAP summarized confidence cutoffs rather than one cutoff.

Validation selection

The sweep compared 24 prediction settings without retraining: four confidence cutoffs, three duplicate-box overlap thresholds, and filtering within or across identities. Validation F1, which combined precision and recall, selected a confidence cutoff of 0.50 and overlap threshold of 0.70, with filtering within each identity.

Compared with the baseline cutoff of 0.25, false detections fell from 12 to 8, while missed correct detections increased from 7 to 9. The change favoured fewer false alarms at a small cost in recall.

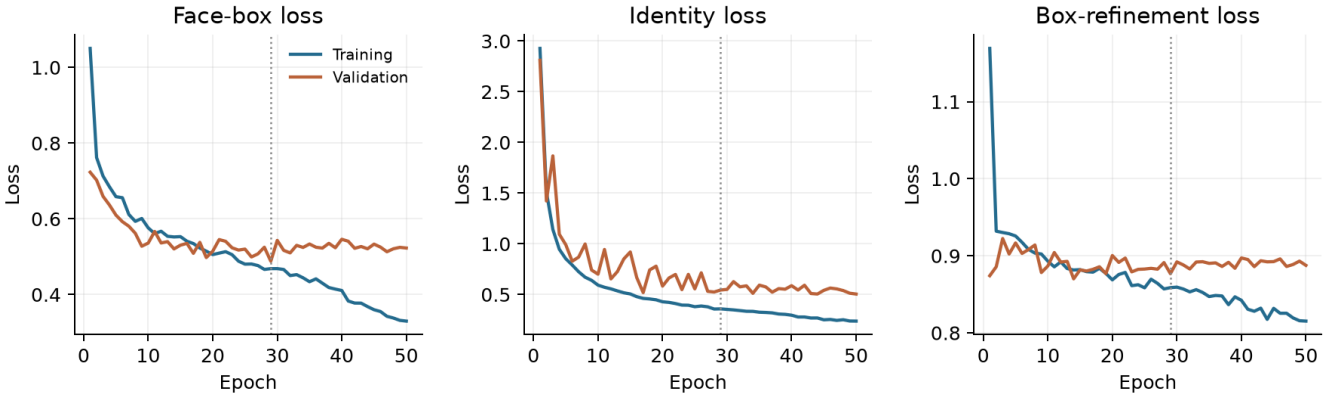


Figure 2. Recorded YOLOv8n training and validation losses. The checkpoint was selected by validation mAP, not the lowest training loss.

Combined pipeline and results

The final pipeline used the saved YOLOv8n and ResNet18 models without retraining. YOLO searched the original image and a smaller view on a plain canvas. Merging these detections helped it find both group-scene faces and close-up portraits. Duplicate and nested boxes were removed before classification.

Each face box was expanded into a portrait crop for ResNet18. The detector scores were normalized and averaged equally with the classifier probabilities. A face kept its box but received an unknown label when the highest combined score was below 0.61. If no box was found, ResNet18 returned a whole-image prediction without localization.

Detection mode, canvas size, model weight and rejection threshold were selected on validation by maximizing correct names minus wrong names. The selected configuration searched both scales. The same pipeline supported the batch command and interactive app.

Test results

Method and input	Correct	Wrong	Unknown	Missed
YOLO only / scenes	104	8	0	1
ResNet18 / scene crops	71	41	0	1
Combined / no rejection / scenes	104	8	0	1
Combined / final / scenes	85	1	26	1
YOLO only / portraits	0	1	0	16
ResNet18 / portraits	11	6	0	0
Combined / no rejection / portraits	16	1	0	0
Combined / final / portraits	16	0	1	0

Scenes contained 113 faces; portraits contained 17 photographs. YOLO alone also produced one extra named box on scenes. ResNet18 used detector-provided crops for scenes and whole images for portraits. Portrait scoring checked the identity of the largest box or fallback prediction; scene scoring also required IoU of at least 0.50.

Rejection reduced wrong scene identities from eight to one, but also withheld 19 previously correct labels. Precision among named faces reached 98.8%, while the share of all faces correctly named fell to 75.2%. Without rejection, combining scores did not improve scene accuracy. The two-scale pipeline correctly named 16 portraits, compared with 11 for ResNet18 on whole photographs.

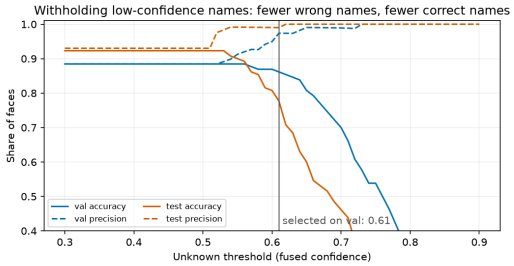


Figure 3. Rejection traded fewer wrong names for more withheld names. The threshold was selected on validation; test curves were descriptive.

Final gallery and identity errors

The repository contained the gallery examples listed below. It was a collection of illustrated results, separate from the test set used to calculate scores. The 20 correct portraits comprised 16 test photos and four validation photos, with their splits labelled.

Gallery requirement	Included in the repository
At least 20 correct single-face examples	20 correct portraits
At least 5 misclassified examples with analysis	6 wrong-label examples, each with analysis
At least 10 multi-face detection examples	10 scenes with bounding-box overlays

The six wrong-label examples included two known identities and four TA-grid faces assessed by the team as unfamiliar. An additional unknown rejection was shown separately from wrong labels. TA assessments were illustrative, not supplied ground-truth labels.

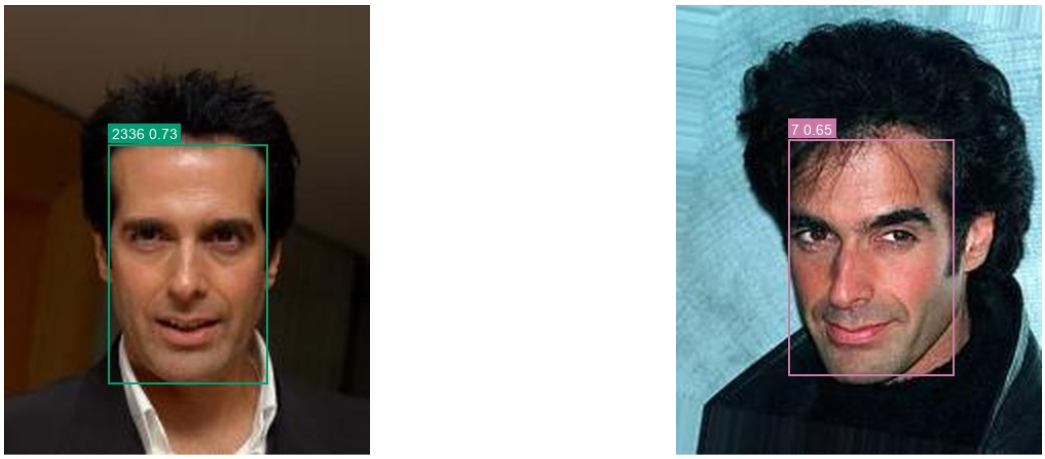


Figure 4. Left: a correctly identified test portrait of 2336. Right: a validation portrait of the same identity was labelled as 7 despite disagreement between the two models. Validation examples illustrated errors but were not independent test evidence.

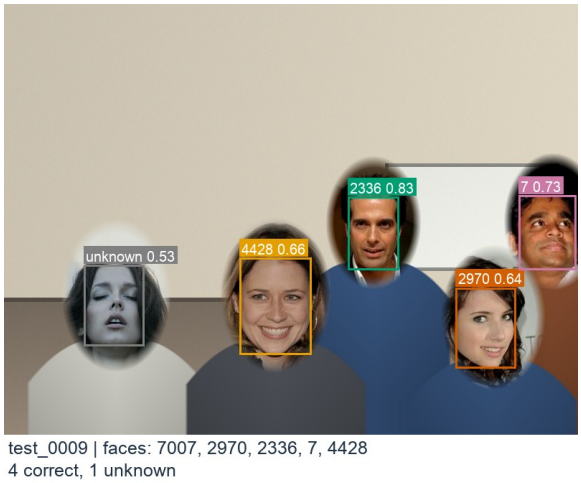


Figure 5. Four faces were named correctly and one was withheld in this test scene. Grey indicated uncertainty, not a confirmed unfamiliar person.

The full results gallery contains the individual images, short error explanations and ten annotated multi-face scenes.

Analysis and limitations

What improved and what remained difficult

The final system combined both models in one command or app. Searching at two scales addressed close-up faces that the original detector often missed. Averaging the model scores provided a way to withhold uncertain labels, but did not make every identity more accurate.

On the scene test, identities 7007 and 7 accounted for most withheld labels. This reduced wrong names while leaving many known faces unidentified. The high precision described only faces that received a name; it did not mean nearly every face was identified correctly.

Limits of the evaluation

The final pipeline was tested on 17 individual portraits and 36 synthetic group scenes. It correctly named 16 portraits and withheld one. The gallery added four correct validation portraits to reach the required 20 examples; these additions did not change the test scores. The group scenes reused the test portraits in different layouts, so they tested different arrangements rather than an entirely new collection of photographs.

Splitting source photographs before composition prevented direct photo reuse across training and test scenes. However, the scenes still varied only a small source collection on drawn backgrounds. YuNet generated the reference boxes, so box scores measured agreement with automatic annotations rather than independently verified ground truth.

An earlier test and gallery review revealed close-up detection failures. The pipeline was then revised to search both scales and remove nested boxes, and its settings were selected again on validation. The final test was therefore not a single blind evaluation of an unchanged design. Portrait-crop framing was also estimated from all 121 source photographs, including evaluation photographs. A stronger follow-up would fix these choices using training and validation data before testing on new images.

Unfamiliar faces and reproduction

The TA grid illustrated a remaining limitation: the final pipeline withheld two faces the team treated as unfamiliar, but assigned known IDs to four others. Both models could confidently agree on a wrong identity. The unknown label was therefore a confidence-based rejection, not a reliable unfamiliar-person detector.

Running the committed settings on Windows reproduced the final reported counts. Repeating the full validation search selected a different detection mode on that setup. The report used the saved configuration; the search was sensitive enough that its outcome should not be assumed identical across environments.

Classification accuracy measured identity predictions on individual photos. Detection mAP also assessed box placement across scoring thresholds. The final counts separated correct, wrong, unknown and missed faces. These results answered different questions, so a larger number in one table did not necessarily mean a better model. More independent real photographs would be needed to assess broader reliability.

Ethical considerations

This project examined a small facial-recognition system in an academic setting. Its results showed that finding a face, assigning the right identity and deciding whether a label should be trusted were different problems. Technical accuracy alone did not establish fairness or suitability for real-world use.

Bias and fairness

Only five identities were represented. Augmentation produced more scenes, but did not add new people or a wider range of appearances. Performance was not measured across age, gender or skin tone, so the results did not establish equal accuracy across demographic groups.

Errors were also uneven within the selected identities. The final pipeline withheld many faces belonging to 7007 and 7 while identifying other identities more consistently. Reporting only precision among named faces would hide that difference. Broader evaluation would need representative photographs and separate checks for groups affected by errors.

Privacy and consent

CelebA photographs were publicly available, but that did not establish consent for every facial-recognition use. An academic project still needed to consider how images were shared and whether people could reasonably expect their faces to be used for identification.

Facial images can identify people and cannot be replaced like a password. A broader application would need a clear purpose, suitable consent, restricted access and limits on retention. Unnecessary image collection, including storage of webcam inputs, would increase privacy risks without necessarily improving the task.

Misuse and mistaken identification

The same technology could support surveillance or tracking without a person knowing. Wrong labels could also cause harm if treated as verified identities in security or other consequential decisions.

The unknown option reduced some wrong labels, but it was not a reliable test for unfamiliar people. In the team-assessed TA grid, some unfamiliar faces still received confident known IDs. Agreement between both models did not make those labels trustworthy.

The system was therefore appropriate as an educational demonstration, not as evidence of identity. Any broader use would need stronger testing, data protections and human review. Its limitations should be shown alongside successful examples rather than left out of the presentation.